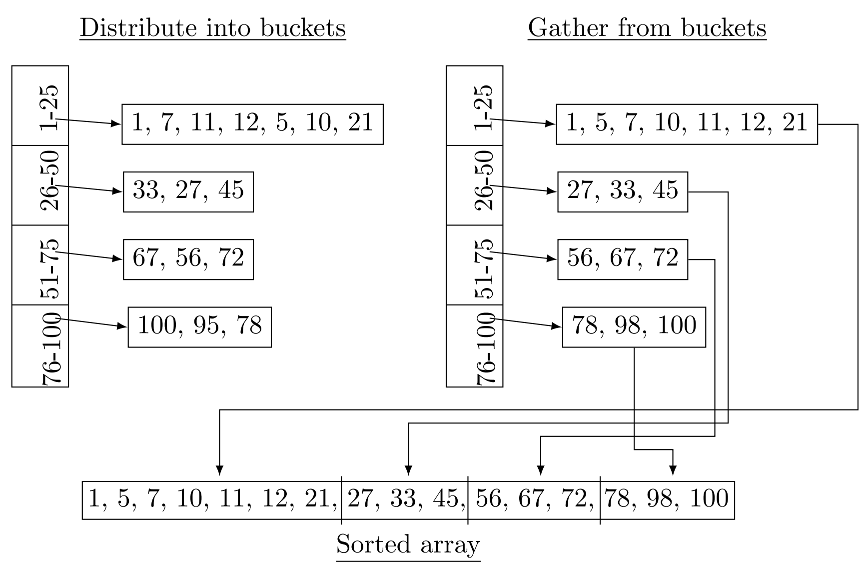
Bucket sort

Bucket sort distributes the set of input elements into a fixed number of groups. Suppose we use k groups. Each element is mapped to a group called a bucket. Essentially, a buckte is a storage for the associated group. So, one element belongs to a unique group. We define the bucket index of an element by a range. The method of partitioning the elements may be likened to a chain hashing scheme. We sort each bucket of elements using one of sorting scheme sucha as insertion sort, bubble sort, quick sort, or any other stable sorting algorithm. Then we gather the elements iterating over the buckets in bucket order and place them in the original array. Let us now summarize the major steps:

- Define k buckets by specifying a distinct range for each bucket.
- Distribute the input elements into the buckets by range index mapping.
- Sort the elements in each bucket using any stable sorting.
- Gather the elements in each bucket iterating over the bucket in their index order.

To illustrate this concept more clearly, let's consider an example.



The logic of bucket sort depends only on the specification of elements to correct mapping of bucket index. It involves a simple table lookup to determine the index of bucket to which an element maps. Furthermore, we can sort the elements in a bucket concurrently with those in other buckets. The concurrent execution of bucket sort can be lot more faster than other sequential sorting algorithms.